The Interview-Compatible Recommender: A Measuring Instrument for Cold-Start Conversational Recommendation

Thesis Chapter A (v2)

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Abstract

Before one can study *how* to interview a cold-start user—which questions to ask, in what order—one needs a recommender that can *be* interviewed: a model that folds an arbitrary-length, mixed-channel set of answers into its belief and re-ranks a large catalogue, without per-user retraining and without trading away full-profile accuracy. We argue that this recommender is a *measuring instrument* for the entire elicitation-policy literature, and that its requirements are more demanding than any single published system meets. The demanding part is *ingestion*: folding an arbitrary-length set of mixed-type answers into a recommender that is already at the accuracy frontier. We state that as five requirements—**R1** state-of-the-art full-profile accuracy, **R2** arbitrary-length cold-start fold-in, **R3** channel-agnostic input (items, out-of-catalogue concepts, entities and continuous directions through one interface), **R4** a queryable belief over the user, and **R5** monotonicity (a truthful answer never lowers ranking quality)—and an acceptance battery (per-build gates **G0**–**G9** plus one-time certifications **C1**–**C2**) that operationalises them. R2 and R3 carry the weight: R4 is what the instrument *provides* to a policy, and what a policy should do with it is the companion chapter's subject. A survey of six model families shows the literature is *bifurcated*: the models that hit the full-profile bar (shallow linear/autoencoder recommenders such as EASE [40] and RecVAE [38]) are item-only point estimators, while the models with an uncertainty-native, multi-channel belief (soft-attribute Gaussian elicitation [5], unified item+attribute bandits [23], conjugate critique folds [44]) do not report full-profile accuracy at the R1 bar. We are aware of no published system that occupies the conjunction of all five. The instrument that closes this gap is a *frozen* certified shallow-SOTA tower over whose latent every channel is folded as a direction paired with a value, with an analytic conjugate-Gaussian belief maintained over the same latent for a policy to query; three of its input legs—signed/graded values, out-of-catalogue concepts, and continuous direction tokens—are architecturally inexpressible in the item-indicator basis of the R1-bar models. This chapter contributes the requirement framework, the taxonomy and gap analysis, the instrument's design, and the experimental protocol (a metric-certification bridge to the published ML-20M numbers, then G0/G1/G2 on the ML-25M harness). Our in-house RecVAE realisation sets the bar at full/tail NDCG@10 0.3540/0.2497 on the canonical ML-25M protocol, and the graded-native interview tower now *meets* it: a single certified training run reaches 0.3482/0.2462 on the held-out test cohort—a paired-bootstrap tie with the bar—while reproducing the frozen decoder's ranking exactly at the empty interview, so G0 (full strength preserved) is passed rather than assumed. The remaining per-build gates have all been run and pass. With the tower fixed, the instrument makes a question the elicitation literature has largely left implicit directly measurable: under one strategy-free selection criterion, a cold user answers 2.18 of eight item questions but 6.02 of eight concept questions—a factor of 2.8—and the concept channel leads on the opening of an interview and on tail accuracy throughout.

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1 Introduction: the recommender as a measuring instrument

A *cold-start* conversational recommender must elicit taste from a short interview and then rank a large catalogue. Two questions are entangled: *what to ask* (the elicitation policy) and *how to turn answers into recommendations* (the recommender). The elicitation-policy literature is large and active, but almost every comparison in it is confounded: a policy's apparent value depends on the recommender it drives, and reported gains routinely conflate a better *question-selection strategy* with a better *answer-ingestion model*. Before we can measure elicitation strategies in isolation, we need to fix the recommender—and not just any recommender, but one strong and flexible enough that a good interview is not wasted on it. We call such a recommender an *interview-compatible instrument*.

Five requirements. An instrument fit for measuring conversational elicitation must simultaneously satisfy:

R1 — full-profile SOTA. On a full interaction history it must rank at the level of the strongest shallow collaborative-filtering baselines (EASE [40], RecVAE [38]), so that elicitation is measured against a recommender nobody can dismiss as weak [12].

R2 — any-length cold-start fold-in. It must ingest an evidence set of *any* cardinality, from 0 (fully cold) upward, without per-user retraining.

R3 — channel-agnostic input. Items, out-of-catalogue concepts, entities, and free text (and, for the continuous policy, direction tokens) must all fold through *one* interface as arbitrary embeddings, not a fixed categorical schema.

R4 — queryable belief. It must expose an explicit covariance over the user belief, not only a point estimate, so that a policy *can* act on what is not yet known. The instrument's obligation is to provide and verify that belief; whether and how a policy exploits it is a separate question.

R5 — monotonicity. A truthful answer must never *lower* ranking quality; adding evidence should, in expectation, only help.

The bifurcation observation. The central empirical fact this chapter documents is that R1 and {R3, R4, R5} are *anti-correlated across the published literature.* The models that hit the full-profile bar—closed-form linear autoencoders (EASE [40], EDLAE [41]), amortised VAEs (Mult-VAE [25], RecVAE [38]), and training-free graph filters (GF-CF [37], BSPM [8], Turbo-CF [32])—are *item-only point estimators* on a one-hot indicator basis. The models with an uncertainty-native, multi-channel belief—soft-attribute Gaussian elicitation [5], item+attribute Thompson bandits [23], conjugate critique folds over a frozen factorisation [44], and per-item Bayesian NL elicitation [3]—do *not* report full-profile accuracy at the R1 bar. We call these the *accuracy corner* and the *belief corner.* No published system we are aware of occupies both (§4).

Contributions. This chapter is the instrument paper of the thesis. It does not claim a new elicitation policy (that is the companion method chapter); it claims the *recommender that makes policy measurement possible.* Concretely:

1. **One latent, many channels, at the accuracy frontier.** A frozen tower that ties the strongest shallow baseline on a shared ruler, over whose latent every channel is folded as a direction paired with a value: item factors, out-of-catalogue concept directions, and directions produced outside the catalogue entirely. The unification is of the *representation*—one space, one form of observation—not of the code path. We frame R1∧R2∧R3∧R4∧R5 as a single requirement no published system meets, and give a six-family taxonomy and per-near-miss gap analysis (§3, §4) locating which requirement each candidate fails.
2. **Three architecturally-inexpressible legs.** We isolate the three input channels that are impossible (not merely untuned) in the item-indicator basis of every R1-bar model: (i) signed/graded per-entity values, (ii) out-of-catalogue concepts, and (iii) continuous direction tokens. These are the narrow, defensible novelty legs of the channel-agnostic interface (§4, §5).
3. **The instrument.** A design for a *frozen* certified shallow-SOTA tower with an analytic conjugate-Gaussian precision-accumulator belief layer over its latent, into which every channel folds as a linear observation (§5). The frozen-tower+exact-conjugacy shape is the one mechanism with a principled monotonicity story [44]. Freezing is *not* what separates us from the soft-attribute line—Göpfert et al. [16] and Bıyık et al. [5] both freeze the collaborative model and learn attribute directions separately, and say so explicitly. What separates us is an *open* concept vocabulary carrying no per-concept supervision, at the R1 accuracy bar (§4).
4. **An acceptance battery as a reusable certificate.** We define three headline outcome gates—full-strength preserved (G0), posterior strictly shrinks per question (G1), and cold-start per-question NDCG rises on *full and tail* versus random (G2)—and, because these outcome gates admit several false positives (a sign-blind fold, a circular answer model, a decorative covariance), extend them into a full battery of per-build gates G0–G9 plus one-time certifications C1–C2 that

add the semantic, artifact-control and answer-model-transfer checks. We propose the battery as a portable certificate any interview-compatible recommender should pass (§2, §7).

We are careful throughout to distinguish what is *new* (the conjunction; the frozen-certified-tower wedge; the acceptance battery on a SOTA tower) from what is *pre-empted* and must be cited (attributes-as-directions plus a Bayesian fold [5]; unified item+attribute belief [23]; conjugate folds over a frozen factorisation [44]; permutation-invariant set-encoders with information-gain acquisition [30]; value-carrying set-input cold-start recommenders [26]; the non-degradation-of-information principle [15, 6]). "Uncertainty-native recommender" alone is a crowded 2024–25 area [7] and is *not* claimed as novel.

2 Requirements and gates

We now state the requirements and gates formally. Let the catalogue have n items and let the frozen tower expose a d-dimensional latent in which each item i has a factor row $\mathbf{Q}_i \in \mathbb{R}^d$. An *evidence set* after t questions is a set of observations $\mathcal{E}_t = \{(\boldsymbol{\phi}_j, y_j)\}_{j=1}^{t}$, where $\boldsymbol{\phi}_j \in \mathbb{R}^d$ is the latent direction of the queried entity (an item factor, a concept centroid, an entity embedding, or a continuous direction token) and $y_j \in \mathbb{R}$ is the (possibly graded, possibly signed) answer. The instrument maintains a belief $b(\mathcal{E}_t) = (\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$ over the user latent $\mathbf{u} \in \mathbb{R}^d$ and scores item i by a fixed read-out $s_i(\boldsymbol{\mu}_t)$ of the base tower.

Requirements (formal).

R1. With $\mathcal{E}$ equal to a user's full history, $\text{NDCG@10}\big(s(\boldsymbol{\mu}(\mathcal{E}))\big)$ ties the strongest shallow baseline on a shared harness (Def. 1).

R2. $b(\mathcal{E}_t)$ is defined and computable for every $t \geq 0$ with no gradient step at inference; t=0 yields the prior belief (a non-personalised ranker).

R3. The update from $\mathcal{E}_t$ to $\mathcal{E}_{t+1}$ depends on the new observation only through the pair $(\boldsymbol{\phi}_{t+1}, y_{t+1}) \in \mathbb{R}^d \times \mathbb{R}$. No channel supplies anything else and none is privileged by type, so any entity with an embedding is a legal observation—including entities absent from the catalogue. The requirement is on the *observation*, not on the number of code paths that consume it.

R4. $\boldsymbol{\Sigma}_t$ is an explicit, queryable covariance, correct in the sense of G1a/G1b, and readable by a policy for any direction w as $w^\top \boldsymbol{\Sigma}_t w$.

R5. For a truthful answer y_{t+1}, $\mathbb{E}\big[\text{NDCG@10}(s(\boldsymbol{\mu}_{t+1}))\big] \geq \text{NDCG@10}(s(\boldsymbol{\mu}_t))$. We treat R5 as an *empirical* property to be demonstrated per step, not a theorem for a learned ranker (§8).

Definition 1 (Ties SOTA)**.** *A recommender* ties *a baseline on a metric if their scores are within the paired per-user bootstrap confidence interval on one shared harness (same users, items, split, metric depth). A number computed on a different metric/split (e.g. NDCG@10 on ML-25M vs. NDCG@100 on the ML-20M strong-generalisation split) is* not *a tie until bridged (§7.2).*

Gates (acceptance battery). The three headline gates below (G0/G1/G2) are *outcome* gates: they certify that the full-profile strength is preserved, that the posterior is usable, and that the cold-start curve rises. On their own they are insufficient (see below), so the instrument is accepted only against the full battery—per-build gates G0–G9 (every build, all cheap) and two one-time certifications C1–C2. Table 1 is the compact reference; the definitions follow.

Per-build gates.

G0 — full strength & identity. Evaluating on the belief mean with the full profile reproduces the frozen tower's full-profile NDCG@10 (full *and* tail)—a paired-bootstrap tie, or bit-identity under the frozen-fold design—so the belief layer adds no full-profile regression (certifies R1 survives).

G1a — posterior shrinks. The posterior covariance shrinks in expectation over answers, with *refusals exempt* (a correct model does not shrink on a refusal).

G1b — shrinkage is directional. The shrinkage is at least twice as large along the queried $\boldsymbol{\phi}_{t+1}$ as elsewhere. G1a and G1b together certify R4: the covariance carries per-direction structure, not merely a global scale.

G2 — cold-start curve rises on full and tail. Starting cold, per-question full and tail NDCG@10 must strictly *increase* from the first to the last turn (not merely fail to drop), judged by the *curve/endpoint at budget* rather than every-step significance; plus an out-of-envelope canary that a single true answer from cold, on each channel's best-static opening question, raises both full and tail above the no-answer intercept. This is the operational form of R2$\wedge$R5, and it is a gate on *ingestion*: whether a better question *order* exists is a selection question, which this chapter deliberately does not adjudicate (§6.4).

G3 — sign & intensity fidelity. Negating every answer's sign lowers NDCG (CI excluding zero, per channel); neutralising values to "indifferent" while freezing membership costs NDCG; and quality is monotone in stated intensity, so a binarising ablation must lose.

G4 — refusals are inert. A refusal burns the turn and moves the belief by nothing. In our design this is architectural rather than fitted—an unanswerable question is never folded—so the property holds by construction.

G5 — concept/channel specificity. Folding a concept ranks its member items above matched non-members (member-lift AUC > 0.8) *and* beats that concept's own popularity projection (top principal component partialled out), with a dedicated channel at least matching a trained member-bag injection.

G6 — existential controls. Wrong-user, shuffled, and placebo-constant answers through the identical pipeline *gain nothing* relative to the true-answer arm (a control may legitimately fall *below* the no-answer intercept—a stranger's taste is worse than no information, not equivalent to it), and duplicating an answer changes nothing (the belief responds to information, not cardinality).

G8 — $\boldsymbol{\Sigma}$ load-bearing & set coherence. Replacing $\boldsymbol{\Sigma}$ with an isotropic $c\mathbf{I}$ *degrades* the acceptance-deciding number (else the R4 claim is withdrawn), and the belief and ranking are invariant to answer order and to conflicting re-asks.

G9 — answer-ingestion isolation. On a *fixed*, user-independent question set the curve still rises with t from answer content alone, separating instrument *ingestion* from question-selection *policy*.

One-time certifications (per instrument, not per build):

C1 — published-number bridge. The implementations snap to the published full-profile numbers on the canonical split (*met, with one disqualification*: EASE and RecVAE reproduce to the third decimal on the ML-20M Liang split and Mult-VAE near-snaps at -0.004, while Mult-DAE fails at -0.022 and is dropped; §7.2).

C2 — answers carry no recommender geometry. Every simulated answer is computed from observed behaviour alone: an item answer is the user's own rating, and a concept answer is a log watch-lift, $\log_2\big((n_c + \frac{1}{2})/(e_c + \frac{1}{2})\big)$, formed from the user's watched members n_c and the count e_c expected from their volume and the concept's catalogue popularity. No latent, no factor, no score and no quantity derived from the recommender enters an answer at any point, and the absence is asserted in code rather than assumed. The gain therefore cannot be the instrument recovering its own geometry, because that geometry is nowhere in the input.

Why the outcome gates alone are insufficient. G0/G1/G2 are outcome gates, and several broken instruments pass all three. A *sign-blind* fold that updates its belief from only the *direction* of the queried entity but never the answer *value* still reproduces the frozen tower (G0), still shrinks $\boldsymbol{\Sigma}$ along $\boldsymbol{\phi}$ per question (G1), and still beats a random-question control (G2)—because beating random certifies that a good question was *selected*, not that the answer was *ingested*; only the sign-flip test (G3) and the fixed-question ingestion isolation (G9) expose it. A *circular* answer model that generates answers from the instrument's own latent geometry would inflate the G2 curve with no transferable signal. The risk is not hypothetical in this literature: evaluations commonly grant the simulated user privileged access to the target—PEBOL's simulator answers from the ground-truth item description [3], and unified item-and-attribute CRS work assumes the user holds and reveals clear preferences over every attribute and item [10]. C2 closes this by construction rather than by measurement (§2). And a *decorative* covariance that is maintained but never enters the acceptance-deciding number passes G0–G2 trivially; only replacing $\boldsymbol{\Sigma}$ with an isotropic matrix and demanding a degradation (G8) shows whether the uncertainty is load-bearing. The battery's semantic and artifact-control gates (G3–G9) and the transfer certification (C2) exist precisely to close these outcome-gate false positives.

We report *both* full and tail NDCG@10 for every gate (a project reporting rule: full-only or tail-only hides the effect, which is typically much larger on the tail than on the full catalogue). Every per-build gate has been run and passes (Table 8); G0 is reported in full in §7.

3 Related work: a six-family taxonomy

We organise interview-relevant recommenders into six families and score each against R1–R5 (Table 2). The reading is consistent across families: no family is strong on both the accuracy axis (R1) and the belief-channel axis (R3–R5). Figure 1 plots the same fact quantitatively, with measured full-profile accuracy on one axis. Table 2 is the compact conceptual overview; the chapter's anchor is the per-system *master table* (Table 3), which classifies every reasonable candidate against R1–R5 *and*, where one exists, prints its full/tail NDCG@10 on our shared ML-25M ruler—so that the bifurcation is not merely asserted from published claims but is visible in the one column of measured numbers.

No dash is an unexplained omission. Every unmeasured row carries one of three dispositions. (Systems with neither a measurable number nor a distinct capability profile—e.g. SLIM, strictly dominated by its closed-form successor EASE in every published comparison—are cited in the text but given no row.) *Not run here*: GF-CF, BSPM, the sequence models and EDDI have public code and are runnable on this ruler, but carry no number in this chapter; the graph-filter family is represented by the measured Turbo-CF row. *Best-effort measured*: for systems whose full-profile core is runnable but whose exact recipe is unpublished or benchmark-bound, we report a best-effort replication of that core with every substitution stated—the node recommender that is the Golbandi tree's core, at 0.3065/0.1895, and the RBMF/Functional-MF seed at 0.2534/0.1764. For Bıyık and ConTS we print a dash rather than a number, because the honest statement is that neither reports full-catalogue accuracy at all—Bıyık's NDCG is agreement between a user's true and estimated top-$|S|$ items over 16 sampled users on slates of 5—and measuring our own reconstruction of a mechanism they share would produce a number that is ours, not theirs. That mechanism is measured in §5, where it belongs: as the cost of ranking on a conjugate posterior mean, which is a decision about *our* architecture. The belief corner is blank in this column not because its systems are weak but because full-catalogue top-N accuracy is not what they set out to report—which is precisely the bifurcation. *Not runnable by construction*: PEBOL maintains per-item Beta posteriors over an LLM-scored shortlist of $\sim$100 items and contains no full-catalogue ranker to run; GATE elicits but does not rank; zero-shot LLM ranking of an 18k catalogue is neither computationally nor methodologically meaningful at full-profile scale (and its collaborative weakness is already documented [17]); BCIE requires its authors' knowledge graph, and a port would substitute so much that the number would no longer be theirs—its conjugate-fold core is covered by the Bıyık-style best-effort arm. ICF is the one row whose full-profile core is *already* measured: its underlying model is Bayesian probabilistic MF, so its full-profile ceiling is the measured iALS row (0.2442).

The measured-accuracy column falls as you enter the belief corner. Read Table 3 down its measured-accuracy column. Every *competitive* number—a full/tail NDCG@10 at or near the frontier on our shared ML-25M ruler—sits in block (a) or (b): the item-only linear and amortised recommenders (RecVAE 0.3540, EASE 0.3476, EDLAE 0.3433). The belief corner, block (c), is either *blank* or *measurably weak*. Both of its runnable cores land well below the frontier—the Golbandi node recommender at 0.3065/0.1895 and the RBMF/Functional-MF seed at 0.2534/0.1764—and both carry an ✗ on R1. The rest of the corner—EDDI, BCIE, UNICORN, PEBOL, GATE, the LLM-zero-shot CRS—has a blank accuracy column entirely. This is not an accident of our harness. Most CRS and elicitation papers never report full-catalogue top-N accuracy at all (they report elicitation RMSE, cold-start MRR/MAP on a shortlist, or critique metrics), and several *cannot produce it by construction*: a per-item Beta posterior (PEBOL) or the decision tree itself (Golbandi, whose node recommender we measure only as a best-effort core) does not induce a ranking over the whole catalogue, and an LLM CRS needs an API call per candidate to score one. Conversely, every system at the frontier is item-only and point-estimate—✗ on R3 and R4. The two blocks are disjoint: no row combines a frontier accuracy number with ✓s across R3–R5. The top-right cell of the gap map—all five requirements met *and* a competitive full-profile number—is occupied only by the target row, which is a design, not a result. That empty conjunction is the chapter's thesis, and the master table is where it is impossible to miss.

The accuracy corner (families ii–iii). On the canonical ML-20M strong-generalisation split (10k validation / 10k test held-out users, NDCG@100), the full-profile frontier has been *flat since 2019–2020* and is held by shallow models, not deep nets: EASE [40] at NDCG@100 0.420, Mult-VAE [25] at 0.426, RecVAE [38] at 0.442 (top of this split), with EDLAE [41] in the same class; SLIM (0.401) and WMF/iALS (0.386) are the classic floors. Dacrema et al. [12] is the binding warning: most deeper "SOTA" gains vanish on this exact split. The graph-filter leaders (GF-CF [37], BSPM [8], Turbo-CF [32]) are the 2023–24 speed+accuracy front on Gowalla/Yelp/Amazon but do not report this ML-20M split, so they are admissible to our accuracy tier only if re-run on our harness. Turbo-CF is, and lands at 0.2622/0.1621—mid-pack rather than frontier at this catalogue size. All of family (ii)/(iii) are item-only: their input is a one-hot (or set-indicator) interaction vector, which *cannot encode* a signed value, an out-of-catalogue concept, or a continuous direction—R3 is an architecture property here, not a tuning gap.

The belief corner (family iv, and vi). Bıyık et al. [5] maintain a multivariate-Gaussian belief over a user embedding and treat both item comparisons and "soft attributes" (e.g. "scarier") as directions (concept-activation vectors) in the item space, selecting queries by expected value of information—the nearest thing to R2+R3+R4 together. ConTS [23] unifies items and attributes as Thompson-sampling arms with a posterior over a cold-start user. BCIE [44] folds critiques into a *frozen* factorisation by exact conjugate-Gaussian updates—the closest published analogue of our belief mechanism. PEBOL [3] keeps a per-item Beta posterior updated by LLM-extracted aspects and NLI scoring, reporting cold-start MRR@10 0.27 versus a 0.17 baseline. None of these reports full-profile accuracy at the EASE/RecVAE bar; each is a belief layer, not a certified tower.

The set-encoder and cold-start-NP precedents (families i, iii). EDDI/Partial-VAE [30] is the permutation-invariant set-encoder over an arbitrary observed subset with information-gain acquisition—exactly R2's architecture, but demonstrated outside recommendation and without a full-profile CF result or an attribute/concept channel. TaNP [26] gives a neural-process, stochastic-process uncertainty over a value-carrying set input for cold-start—the R2+R4 shape, but not at full-profile SOTA nor channel-agnostic. These are cited as the mechanism precedents our design instantiates, not as rivals for the conjunction.

The gap map. Figure 1 places every system on two axes: *interview-compatibility* on the x-axis—a coarse ordinal capability count, the number of **R2–R5** marks a system satisfies in the master table (✓=1, ∼=0.5; R1 is excluded because accuracy is the y-axis)—and *measured full-profile NDCG@10 on our ML-25M ruler* (R1) on the y-axis. The population is bimodal: the accuracy corner collapses onto $x \leq 1.5$, the interview-capable systems sit at $x \geq 2$ with weak or non-existent full-profile accuracy, and no system bridges—the top-right, high compatibility *and* frontier accuracy, is empty.

4 Gap analysis

The verdict. No single published system satisfies all of {R1 SOTA-full-profile ∧ R2 any-length set ∧ R3 channel-agnostic via arbitrary embeddings ∧ R4 uncertainty-native ∧ R5 monotone}. The binding near-miss is Bıyık et al. [5]: it fails R1 (a d=50 ALS two-tower rather than a certified RecVAE-class tower, and it reports no full-catalogue accuracy—its NDCG is computed between a user's true top-$|S|$ items and the top-$|S|$ estimated from the posterior, over 16 sampled test users on slates of 5, an in-simulator alignment measure) and gives only an approximate, "generally not Gaussian" posterior; it fails R3 in the specific sense of §4 (a per-concept supervised probe over a closed tag list, not an open vocabulary). Every other candidate fails R3 or R4 outright, and we are aware of no system that occupies the conjunction.

Where each near-miss falls short.

- **Bıyık et al. 2023** [5] — has R2 + R3 (items and soft attributes as directions) + R4 (Gaussian belief), over a *frozen* item tower—they train the CF model and the attribute directions separately, as we do. *Fails R1*: a d=50 ALS two-tower, and accuracy is reported on their own elicitation task (in-simulator top-$|S|$ agreement over 16 users, slates of 5), not the EASE/RecVAE bar. R3 is *per-attribute supervised directions* over an enumerated tag list—each direction is a logistic regressor fitted from labelled positive/negative items, and they train CAVs for the 164 most-frequent tags—not arbitrary embeddings / open concepts / continuous query tokens. R5 is approximate (sampled EVOI over a non-Gaussian posterior). This is the central threat, and our wedge is exactly (a) an *open* vocabulary through one global adapter with no per-concept supervision, (b) the R1 accuracy bar on a shared full-catalogue ruler, (c) exact conjugacy vs. approximate posterior.

- **ConTS** [23] — R2 + R4 + a unified item/attribute action space. *Fails R3* (a fixed categorical attribute schema; no arbitrary embedding, no continuous or open token) and *R1* (a bandit over arms, not a full-profile encoder).
- **EDDI / Partial-VAE** [30] — R2 + R4 (information-gain) + set input. *Fails R1* (not a recommendation model / no CF SOTA) and *R3* (no attribute or concept channel as demonstrated).
- **RecVAE / EASE** [38, 40] — *R1 yes. Fail R3* (an item-only one-hot / indicator basis—cannot express a concept, a signed value, or a continuous direction) and *R4* (point estimate; RecVAE's latent posterior is item-only and is not used as an actionable belief).
- **PEBOL** [3] — R4 (per-item Beta) + R3 (LLM aspects) + strong cold-start. *Fails R1* (no full-profile CF ranking) and its R4 is *per item*, not a shared user covariance; no monotonicity guarantee.
- **SASRec / BERT4Rec** [20, 42] — R2 (re-encode the sequence). *Fail R3, R4, R5* (item-only, order-sensitive, point estimate, non-monotone—a later token can reorder everything). Token input is trivially satisfied but is *not* value-carrying or mixed-type, so "interview = tokens" does not hold; and BERT4Rec's fold-in advantage carries a replicability caveat [34].

The three inexpressible legs. Three input channels are *architecturally impossible* (not merely untuned) in the item-indicator basis of every R1-bar model: (i) graded/signed per-entity values (the basis has no sign and no magnitude—a "dislike" is not representable as a negative one-hot without changing the likelihood), (ii) out-of-catalogue concepts (there is no column for an entity absent from the catalogue), and (iii) arbitrary continuous direction tokens. Folding these as *linear observations in a frozen latent* is the one leg of the conjunction we find nowhere in the literature.

The distinction that matters is between an *open* and a *closed* channel. An item-indicator model can be extended with a concept—as a pseudo-item column, or an extra input unit—but only by adding a column, and a column is fixed when the model is fitted (§7). Its concept vocabulary is therefore frozen at training time. Our interface takes a direction $\phi \in \mathbb{R}^d$ and a value, so any entity with an embedding is a legal observation at inference: a paraphrase, a concept coined after deployment, or a continuous direction emitted by a policy and snapped to a nearby meaningful axis. This is what we mean by channel-agnostic, and it is a property of the interface rather than a quantity to be tuned.

Pre-empted—cited, never claimed as first. We must credit, and do not claim priority over: attribute-answers-as-directions plus a Bayesian fold [5]; unified item+attribute belief and value-of-information [5, 23]; conjugate-Gaussian folds over a frozen factorisation [44]; permutation-invariant set-encoders with information-gain acquisition [30]; value-carrying set-input cold-start recommenders [26]; the non-degradation-of-information principle [15, 6]; and the now-crowded "uncertainty-native recommender" area [7]. Our defensible novelty is only the *conjunction* of all five requirements in one frozen certified tower, plus the three inexpressible legs and the acceptance battery that certifies them.

5 The instrument

The design follows directly from the gap analysis: use the shallow frozen tower that already holds the accuracy corner (R1, and the flat-since-2019 frontier means no deep-net detour is warranted), and add the one belief mechanism with a principled monotonicity story—exact linear-Gaussian conjugacy—over its frozen latent (R4, R5), with a channel-agnostic linear-observation interface (R3). The tower half is built and certified (§7); the belief layer is derived here, fitted on real data, and gated in §7.3.

5.1 Frozen shallow-SOTA tower

The substrate is a certified shallow recommender—a RecVAE-class amortised encoder or a closed-form autoencoder [38, 40]—trained once on the full catalogue and then *frozen*. It exposes (a) item factor rows $\mathbf{Q}_i \in \mathbb{R}^d$, (b) a decoder read-out $s_i(\mathbf{u}) = \mathbf{Q}_i^\top \mathbf{u} + \beta_i$ with a learned bias β_i, and (c) the encoder that maps a full interaction set to a latent. Freezing is the guarantee that G0 (full strength preserved) is a property of the substrate, not of the belief layer. It is a design choice we share with the soft-attribute line rather than one that distinguishes us: Göpfert et al. are explicit that "we do not use the tag data when training the collaborative filtering model" [16], and Bıyık et al. train the CF model and the attribute directions separately [5]. We cite that convergence as independent support for the frozen-substrate design. We deliberately do *not* extend the linear basis with concept columns (that stays in the item-indicator world and fails R3), nor learn the uncertainty head (no monotonicity story, and retraining risks G0). The tower is now *selected and certified*: the graded-native set-encoder T2′, trained once in its pre-registered configuration, scores 0.3482/0.2462 full/tail on the held-out test cohort—a paired-bootstrap tie with the 0.3540/0.2497 RecVAE bar—and reproduces the frozen decoder exactly at the empty interview (§7). It is this checkpoint that the belief layer below is placed over.

5.2 Analytic conjugate-Gaussian belief layer

Over the frozen latent we place a Gaussian belief on the user vector, $\mathbf{u} \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$, and model each answer as a noisy linear read-out along the queried direction:

$$y_j \;=\; \boldsymbol{\phi}_j^\top \mathbf{u} + \varepsilon_j, \qquad \varepsilon_j \sim \mathcal{N}(0, \sigma_j^2). \tag{1}$$

Because the observation is linear-Gaussian, the posterior after $\mathcal{E}_t$ is Gaussian in closed form; maintaining it in *precision* form (a precision accumulator) makes the any-length fold-in a single running sum:

$$\boldsymbol{\Lambda}_t \;=\; \boldsymbol{\Lambda}_0 + \sum_{j=1}^{t} \sigma_j^{-2}\, \boldsymbol{\phi}_j \boldsymbol{\phi}_j^\top, \qquad \boldsymbol{\Lambda}_t\, \boldsymbol{\mu}_t \;=\; \boldsymbol{\Lambda}_0 \boldsymbol{\mu}_0 + \sum_{j=1}^{t} \sigma_j^{-2}\, y_j\, \boldsymbol{\phi}_j, \qquad \boldsymbol{\Sigma}_t = \boldsymbol{\Lambda}_t^{-1}. \tag{2}$$

Equation (2) is exactly the mechanism BCIE [44] uses for critiques over a frozen factorisation; our contribution is doing it over a *certified full-profile SOTA* tower with channel-agnostic tokens, and certifying the G0–G2 protocol on it. Three properties fall out by construction: it is a permutation-invariant *set* operation defined for every $t \geq 0$ (R2); each update adds a rank-1 term along $\boldsymbol{\phi}_{t+1}$, so $\boldsymbol{\Lambda}_t$ only grows and $\det \boldsymbol{\Sigma}_t$ only shrinks (G1); and conjugate updates are monotone in *expected utility* [15, 6]. The last is why R5 is plausible, but—because the tower's dot-product ranker does not act optimally under the belief—per-question NDCG monotonicity is *not* automatic and must be shown empirically (§8).

One answer tightens a whole region, and that is where coarse-to-fine comes from. Because each update adds a rank-one term, the Sherman–Morrison identity gives the effect of asking $\boldsymbol{\phi}$ on the uncertainty along *any other* direction w in closed form:

$$w^\top \boldsymbol{\Sigma}_t w \;-\; w^\top \boldsymbol{\Sigma}_{t+1} w \;=\; \frac{\sigma^{-2}\left(w^\top \boldsymbol{\Sigma}_t \boldsymbol{\phi}\right)^2}{1+\sigma^{-2}\,\boldsymbol{\phi}^\top \boldsymbol{\Sigma}_t \boldsymbol{\phi}}. \tag{3}$$

The reduction is proportional to $(w^\top \boldsymbol{\Sigma}_t \boldsymbol{\phi})^2$—the squared covariance between w and the queried direction under the current belief. A question therefore does not sharpen one axis; it sharpens a whole *correlated region*, in proportion to how strongly that region moves with what was asked. The consequence for a mixed question bank is immediate and geometric rather than heuristic: a concept direction is by construction close to many item directions at once, so folding it collapses a large region per question, whereas an idiosyncratic item collapses little beyond itself. Breadth-before-depth is thus a property of the covariance geometry, not a scheduling rule imposed from outside—and it is what we observe when a purely accuracy-driven greedy is allowed to choose freely from items and concepts together (§6.4). Coarse-to-fine itself is long known in the interview literature [14, 36]; what Eq. (3) adds is an account of *why* it holds in a latent-factor instrument, and a way to compute it for any channel.

The belief separates cleanly into an answer-independent and an answer-dependent part. A property of Eq. (2) deserves to be stated explicitly, because anyone building elicitation on this instrument needs it. The precision $\boldsymbol{\Lambda}_t$ accumulates $\sigma_j^{-2}\boldsymbol{\phi}_j\boldsymbol{\phi}_j^\top$ terms: it depends on *which* directions were queried and on the noise assigned to each, but *not on the answers themselves.* The covariance $\boldsymbol{\Sigma}_t$ is therefore the same for two users who were asked the same questions and replied differently. The mean is not: $\boldsymbol{\mu}_t$ carries $\sum_j \sigma_j^{-2} y_j \boldsymbol{\phi}_j$ and moves with every y_j.

This asymmetry is informative for elicitation. A question-selection rule that reads only the covariance is, by construction, answer-independent—it produces the same sequence for every user, which is the linear-Gaussian, variance-only case in which adaptive selection provably gains nothing over a fixed schedule [21, 19]. A rule that reads the mean, or any quantity derived from it such as the current ranking, is not. Our instrument exposes both, so the choice is available to a policy designer rather than forced by the substrate; which rule to use, and what it is worth, is the companion chapter's question.

One qualification, which the refusal band supplies. The per-observation noise σ_j is not constant across answer types: an unanswerable question is not folded at all, so whether a user *could* answer changes which terms enter $\boldsymbol{\Lambda}_t$. The covariance is consequently answer-dependent to the extent that answerability varies across users—and it varies a great deal, since items and concepts are answered at very different rates (§6.4). Answerability thus enters the belief itself, not only the accounting of a wasted turn.

What the belief layer provides. The layer is built and fitted: $\boldsymbol{\Lambda}_0, \boldsymbol{\mu}_0$ are taken from the tower, and the per-channel noise reduces to eight fitted scalars (a 2×3 channel×confidence precision grid plus a prior scale and a variance floor), calibrated on real data with a pre-fit sign proof ($G = 2291$ over 414,265 concept triples). It passes G1a, G1b, G4, G8 and G9 (Table 8): the posterior shrinks monotonically over answers, shrinks 11.5× more along the queried direction than elsewhere, is order-invariant to 5×10^{-7}, and is unmoved by a refusal. Uncertainty along any direction w—an item, a concept, a phrase—is read off as $w^\top \boldsymbol{\Sigma}_t w$.

That is the instrument's obligation discharged: a belief that is correct, queryable for any channel, and available to a policy. What a policy should do with it—and in particular whether reading the covariance is a good way to choose questions—is a separate question, and the companion chapter's subject.

5.3 Channel-agnostic tokens: one representation, one form of observation

Every channel supplies the same kind of thing—a direction $\boldsymbol{\phi}_j \in \mathbb{R}^d$ in the frozen latent and a value $y_j \in \mathbb{R}$—and it is this that makes the channel set open (R3). Two folds consume those pairs in the system we report: the tower's own set encoder for item evidence, and a trained gated operator for concept evidence (§6.2); the conjugate update of Eq. (2) maintains the belief over the same latent alongside them. We are explicit that these are distinct operators rather than one code path, because the unification that matters is of the representation: any entity with an embedding is expressible as an observation, which is what no item-indicator model can say.

Why the belief is maintained alongside the ranking rather than inside it. The conjugate posterior mean is itself a legal ranker—score item i by $\mathbf{Q}_i^\top \boldsymbol{\mu}_t + \beta_i$—and it would make Eq. (2) the single fold for every channel. We measured that option rather than assuming it: a conjugate-Gaussian belief over a frozen factorisation, ranked by its posterior mean, reaches full/tail NDCG@10 0.2019/0.1200 on our ruler, against 0.3482/0.2462 for the learned fold (Table 7). Ranking on the posterior mean therefore costs roughly 0.15 full NDCG—it forfeits R1 to buy uniformity—and that is why the belief is carried beside the ranking path and read by a policy, not used to produce the rankings we report.

The channels, then, are:

- **Items.** $\boldsymbol{\phi}_j = \mathbf{Q}_i$ (the frozen item factor). A graded or signed answer sets y_j; a dislike is a negative y_j along the same direction—representable here precisely because the observation model (1) has a sign and a magnitude, unlike a one-hot indicator.
- **Out-of-catalogue concepts.** $\boldsymbol{\phi}_j$ is a concept direction in the latent (e.g. a centroid of member factors, or a text-adapter image of the concept), folded exactly like an item—no catalogue column required.
- **Entities / free text.** An embedding is projected into the latent by a fixed adapter and folded identically.
- **Continuous direction tokens.** A policy may emit any $\boldsymbol{\phi}_j \in \mathbb{R}^d$ directly (the continuous-action interface the companion chapter uses); the belief update does not distinguish it from a catalogue entity.

The signed/graded, out-of-catalogue, and continuous legs are the three channels inexpressible in the R1-bar basis (§4).

Any direction is a legal observation: an illustration. The signed/graded leg is delivered and measured (G3a/G3b, Figure 2). The concept leg is delivered over the genome vocabulary. The third leg—an *arbitrary* direction, including one for an entity that has no catalogue column and was never seen in training—is a property of the interface rather than of any particular channel: everything downstream of Eq. (2) consumes a pair $(\boldsymbol{\phi}, y)$, so any entity with an embedding is admissible. We illustrate that concretely.

Each of the 1,031 genome concepts has a direction in the latent and a name in English, so those pairs determine a single global linear map from a frozen general-purpose sentence encoder into the frozen latent, in closed form. We fit it on a random 80% of the concepts and ask what happens to the 20% it never saw: for a held-out concept, the direction predicted *from its name alone* recovers the top of that concept's own item ranking more than 300× more often than chance (0.167 top-10 overlap against a 0.0005 random baseline; cosine 0.638 to the true direction). No per-concept probe is fitted, no inventory is consulted, and nothing is retrained—a phrase becomes a direction by one matrix multiplication.

Because the map is defined on any string, it also accepts phrases that are not in our concept vocabulary. To make that meaningful rather than decorative, each phrase below is screened against the vocabulary two ways—no genome tag occurs as a whole-word substring of it, and no concept sits within cosine 0.6 of it—and we print the *closest* concept we actually hold, so the reader can see how far the phrase sits from anything curated. (The screen is not cosmetic: it rejects "revenge served slowly", "loneliness in a big city" and "a detective who drinks too much", because *revenge*, *loneliness* and *detective* are all themselves genome tags.)
The systematic claim is the held-out number above; the table shows what it buys. The last row makes the point plainest: the nearest concept in the entire vocabulary to "movies about grief" is *feel good movie*, tonally its opposite, and the phrase's direction shares none of its top ten items with that concept's curated direction—the phrase is placed by its own meaning, not snapped to the closest thing on a list. The phrases were drawn from a screened pool, and ones whose meaning the genome vocabulary does not carry at all—"set entirely inside one room"—return lists we would not defend.

We present this as an *illustration of what the interface permits*, not as a contribution to mapping language into collaborative latents—a question with an established literature of its own, in which attribute directions are obtained either by per-concept supervised probes [16, 5] or by per-attribute corpus retrieval [4]. Fitting such a map *well* is a research direction this architecture opens rather than one this chapter pursues: text underdetermines collaborative taste, and how much of that gap a better-fitted adapter can close is exactly the kind of question the instrument now makes measurable.

6 The concept channel: derivation, lineage, and deployment currency

The out-of-catalogue concept leg is the most demanding of the three inexpressible channels—it is the one with no catalogue column, no interaction count, and no sign in the item-indicator basis—so we work it out in full. Three things must be made concrete: how a concept answer becomes an observation pair $(\boldsymbol{\phi}_c, y_c)$ for the belief update of Eq. (2); where each piece sits in the published record (so that we claim only the conjunction, never a component); and why this channel is the one that forces the battery to gate what a *deployed* interview elicits and not only what the operator can express. This section is written at derivation level; the concept-channel numbers below are measured on the internal ML-25M interview harness and are reported as such.

6.1 A signed, exposure-corrected affinity estimator

A concept answer must supply the value y_c that Eq. (1) reads along the concept direction. We want that value to be *signed* (a user can dislike "horror," not merely fail to like it), *popularity-corrected* (a concept made of blockbusters must not read as universal affinity), *volume-corrected* (a heavy watcher must not read as liking everything), and *confidence-bearing* (a concept the user has plausibly never been exposed to is a *refusal*, not a dislike). Two estimators, an implicit and an explicit one, combine to meet this.

Let $\mathcal{I}_u$ be the user's watched set, p_i the catalogue marginal of item i, and let concept c have member set M_c. Write $n_c = |\mathcal{I}_u \cap M_c|$ for the user's watched members and $e_c = |\mathcal{I}_u| \cdot \pi_c$ for the number *expected* under the user's volume times the concept's popularity $\pi_c = \sum_{i \in M_c} p_i$. The implicit *selection* estimator is the add-$\frac{1}{2}$ log watch-lift

$$\mathrm{SEL}(u,c) \;=\; \log_2 \frac{n_c + 0.5}{e_c + 0.5} \;\;\approx\;\; \mathrm{PMI}(u,c), \tag{4}$$

which is precisely the *pointwise mutual information* between the user and the concept: the denominator's π_c divides out concept popularity, and forming e_c from the user's own volume divides out user activity, so SEL is positive for genuine affinity, near zero for popularity-explained co-occurrence, and *negative for avoidance*. Its normalised form is the NPMI, $\mathrm{NPMI} = \mathrm{PMI}/(-\log_2 \hat{p}_{uc}) \in [-1,1]$, which tames PMI's small-count inflation and yields a bounded, comparable score. The explicit *value* estimator is a support-shrunk signed rating residual,

$$\mathrm{VAL}(u,c) \;=\; \frac{n_c}{n_c + \lambda}\,\bar{r}^{\,\mathrm{res}}_{uc}, \qquad \lambda = 3, \qquad \bar{r}^{\,\mathrm{res}}_{uc} = \frac{1}{n_c}\sum_{i \in \mathcal{I}_u \cap M_c} \big(r_{ui} - \bar{r}_i\big), \tag{5}$$

the mean rating the user gives the concept's members *above each item's own mean*, shrunk toward zero by the Beta-style factor $n_c/(n_c+\lambda)$ so that a one-film opinion counts for little. Both estimators are signed and combine into a four-band answer: a *graded like* (SEL/VAL band above the neutral zone), a *neutral* (indistinguishable-from-exposure), a *graded dislike* (band below), and a *refusal* issued exactly when the expected support falls below a threshold, $e_c < \tau$, i.e. when the user has plausibly never been exposed to the concept and no answer should be forced.

Lineage. Every piece is established prior art. The log-lift (4) is PMI/NPMI, the standard popularity-corrected association score. Reading a *magnitude* of implicit consumption as evidence is Hu–Koren–Volinsky's implicit-confidence template [18]—except that HKV lets the count re-enter *only as a loss confidence weight, never as a sign*, which is exactly the assumption our signed estimator breaks. The principled account of why a non-watch is not a dislike is ExpoMF [24]: it separates a latent *exposure* variable (was the user even aware of the item) from *preference*, so that non-consumption is modelled as "not exposed" *or* "exposed-and-disliked"—the generative gold standard our refusal band ($e_c < \tau$) and popularity-corrected e_c approximate cheaply and in closed form. The unbiased-learning route to the same correction is attribute-level exposure propensity [35], which IPS-reweights implicit feedback by a per-attribute propensity; it is the heavier, variance-prone alternative to our moment estimator. Finally, calibrated recommendation [39] builds a user's normalised genre *distribution* $p(g \mid u)$—the closest widely-used "concept profile," but a distribution that sums to one and therefore encodes relative emphasis only: it *cannot express dislike*, which is precisely the expressive gap SEL< 0 fills.

Why the implicit channel, quantified. On an informativeness decomposition (a per-concept linear fit of the oracle concept affinity onto each candidate answer, R^2), the implicit watch-lift dominates: SEL explains 0.376 of the affinity variance, the explicit rating residual VAL a further +0.042 (SEL+VAL 0.398), while an LLM-elicited ordinal answer explains only 0.017 (an internal decomposition)—implicit outperforms the explicit LLM answer by roughly 22×. Because SEL is built from raw watch counts and item marginals, it uses no recommender geometry, so this signal is *non-circular by construction* (it even beats a model-geometry projection of the same residuals). This is the evidence that the concept channel's value is worth carrying signed rather than as a mere membership indicator; the decomposition is an informativeness measurement, not an interview curve, and is reported as such.

6.2 The fold operator: whitened directions and a gated fold-to-point

The direction ϕ_c must point where the concept's taste lives, not where popularity lives. A raw member centroid fails this: a shared popularity component (random concept pairs have cosine ≈ 0.5) masks the genre geometry. Removing it—centre the member factors and strip the top-1 principal component (the popularity axis), the all-but-the-top whitening of Mu and Viswanath [31]—turns the centroid into a usable direction: concept-membership discrimination through the frozen decoder rises from AUC 0.44 (raw, at chance) to 0.94 (whitened). We therefore take ϕ_c to be the *whitened member centroid*, IDF-weighted by $1/\log(1+|M_c|)$ so broad genres do not dominate. Whitening is a fix, not a contribution, and is cited as such.

The operator that consumes (ϕ_c, y_c) is where the design decision lives, and the two candidate families separate cleanly under ablation. An *additive-union* operator scores each item by summing concept contributions on a popularity floor, $s_i = \beta_i + \sum_c w_c \langle \mathbf{W}_d \mathbf{Q}_i, \phi_c \rangle$. It has the right monotonicity floor (an empty interview scores exactly β_i) but the wrong composition: it scores the *union* of directions, so correlated sibling concepts ("dark," "gritty," "noir") re-add the same acclaim axis and over-count. Measured against a no-answer intercept of 0.1279 full NDCG@10, the additive channel peaks early (about +0.011 at two answers) and then *craters* as answers accumulate, reaching 0.0991 at eight answers—below the intercept. A *trained gated fold-to-point* operator fixes this by folding the answers into the latent itself and letting a frozen decoder apply the full collaborative (intersection) structure:

$$\mathbf{u} \;\leftarrow\; \mathbf{u} + \sigma\big(g([\mathbf{u}; \phi_c; y_c])\big) \;\odot\; h([\mathbf{u}; \phi_c; y_c]), \tag{6}$$

with the gate g and the delta MLP h *zero-initialised* (so at initialisation $\mathbf{u}$ is unchanged and G0 survives) and the accumulated shift norm-capped per user. In the concept-operator ablation it is monotone in the number of answers to eight (0.1759/0.0841 full/tail at eight answers, up from 0.1322/0.0503 at one) and, crucially, *redundancy-robust*: feeding it eight *correlated* top-SEL concepts still reaches 0.1685/0.0906, above its four-answer level (0.1572/0.0777), exactly where the additive-union operator degraded. The gated fold is the concept operator we adopt; Eq. (6) is the concept-channel instantiation of the linear-observation fold, with the belief covariance maintained analytically as in §5.

Lineage. The skeleton—an attribute as a latent direction, folded by a Bayesian belief update over a frozen factor space—is well-trodden, and we claim none of it in isolation. Bıyık et al. [5] treat a soft attribute as a concept-activation-vector direction and fold signed answers into a Gaussian belief; M&Ms-VAE folds a rating-posterior and an attribute-posterior over one latent by a product-of-experts merge [1]; its successor adds an independent *signed* critique axis through learned positive/negative conditioning tokens, and reports that negative critiquing saturates after about five turns [2]—a concrete ceiling for any signed-attribute channel, consistent with our own front-loaded concept curves. BK-VAE folds keyphrase-activation-vector directions by a Bayesian critiquing step [29]; the gate/MLP of Eq. (6) is a per-token generalisation of FiLM conditioning [33], which a single global affine cannot do because it would apply the same modulation to item and concept tokens alike. And the decision to *freeze* the tower rather than co-train a concept head is a direct reading of the cautionary result that a bolt-on critique head costs recommendation accuracy [28].

What none of these holds jointly—and what we are aware of no system holding—is the conjunction this channel occupies: *continuous, out-of-catalogue concept directions* (a whitened centroid, or a text-adapter image, with no per-attribute probe to train), driven by a *signed, exposure-corrected implicit affinity* (Eqs. (4)–(5)), folded over a *frozen, certified full-profile SOTA tower.* The components are borrowed; only their conjunction is claimed.

6.3 Deployment currency: why a capability gate is not enough

A gate on what an operator *can* express must be paired with a gate on what the deployed system *does* express, and the concept channel is where the distinction bites. The fold of Eq. (6) is sign-capable: negating every *concept* answer moves NDCG by -0.2012 with a clean CI, so the concept channel passes G3's sign-flip test on its own operator. But that certifies the operator, not the interview. Driven down a realistic, user-independent question bank, most concept answers are weak positives, and an answer model that never emits a negative can accumulate them until the ranking drifts *below* the no-answer intercept while every capability gate still reads green.

We therefore evaluate the channel in *deployment currency*: a fixed, user-independent bank under a per-question budget where every question spends budget whether or not it is answered, scored at the endpoint against the intercept (the G9 ingestion-isolation setting). The acceptance form of the concept channel is not "can the operator express dislike" but "does the realizable interview stay above the no-answer intercept at budget." This is the criterion the signed four-band estimator of §6.1 is built to satisfy, and §6.4 reports it.

6.4 Results: the signed retrain, and where the concept channel wins

The signed retrain of §6.3 has since landed, and we ran the two channels—concepts and items—through *one* deployment-currency harness (10,000 held-out cold users, per-question budget, every question spending budget whether answered or not, scored on full *and* tail NDCG@10 against the no-answer intercept 0.1279/0.0192). They tell a consistent story: the signed answer keeps the deployed interview above the intercept, and the concept channel's value is *shaped*—it owns the opening of an interview and cedes the long tail of it to items—and its largest prize is one we do not yet capture.

The signed answer turns the deployment decline into a gain. Re-running the pre-registered triple gate on the *retrained* model (not merely the eval-only fold of §6.3) clears all three taste criteria. Signed answers beat the positive-only clip by $+0.0104$ full NDCG@10 at the sixteenth question (CI $[+0.0075, +0.0134]$): the clip-up curve sinks below the intercept by the eighth question and stays there (0.1268 at q=16), while the signed curve holds above it (0.1372 in this gate's arm). The counterfeit control—an answer built only from user volume and concept popularity—reproduces *none* of the gain (it declines, -0.045 at q=16), so the signal is taste, not a volume×popularity artifact; and permuting the answer values across concepts collapses the gain ($+0.0335$ CI-excluding-zero relative to the permutation), so it is the answer *values* that carry it, not fold cardinality. The two controls together establish what the channel is doing: the gain is carried by the answer *values*, and it is not a restatement of how much the user has watched.

Both channels carry an interview. The instrument's claim on the elicitation side is deliberately modest, and it is worth separating from the claim we do *not* make. What the instrument chapter needs is that *both* channels fold and both move the ranking off the cold intercept—which is what makes the instrument usable for measuring policies over a mixed question bank. That much is unambiguous: an interview conducted entirely in out-of-catalogue concepts, a channel no R1-bar baseline can express at all, lifts a frontier-accuracy recommender substantially and monotonically above its 0.1279/0.0192 starting point, and so does an interview conducted entirely in items.

What we deliberately do *not* state here is a ranking between the two channels, and the reason is methodological rather than diplomatic. Any such comparison is only as meaningful as the question-selection strategy each channel is given, and pairing one hand-picked heuristic per channel—a popularity-ordered item bank against a polarization-ordered concept bank, say—measures the two heuristics at least as much as it measures the two channels. We therefore compare the channels under a single strategy-free criterion: the *best static question sequence* for NDCG over an items-only bank, a concepts-only bank, and a combined bank of both, with the sequences constructed on the validation cohort and scored on the disjoint test cohort. Every bank meets the same criterion on the same cohorts at the same budgets, and the combined bank *contains* the items-only bank, so what separates the arms is the channel rather than whichever heuristics happened to be paired. All three arms are in Table 5. The answerability figure underneath them is the striking one: across a sixteen-question interview a user answers on average only 3.6 of the item questions, the rest being items they have not seen. That is the structural pressure the concept channel exists to relieve. Three things follow.

Concepts own the opening, on both metrics. At one question the concept bank leads on full (.1436 vs .1405) and on tail (.0323 vs .0275), and the combined bank is *identical* to concepts-only through q=4—given free choice over five hundred items and five hundred concepts, the optimiser spends its first four questions on concepts. The emergent combined order is `c c c c i i i i c i i i i i i i`: concepts first, then items, with one concept returning at position nine. Nothing in the objective encodes "ask broad things first." Coarse-to-fine is long known in the interview literature [14, 36] and we claim no discovery; what is new here is that it emerges unprompted in a *mixed* action space, and that §5 gives a geometric account of why (Eq. (3)).

Concepts dominate the tail at every budget, without being asked to. At q=8 the concept bank reaches .0698 tail against items' .0538, a 30% margin, and the combined bank is the best tail arm at q=16 (.0776 vs .0679, +14%). **The greedy maximised full NDCG@10 only**—the tail was never an objective in any arm—so the tail advantage is a property of the channel, not something the optimiser was steered toward.

Items win full accuracy on the long interview. From q=8 the item bank leads on full (.1863 vs .1776), and at q=16 combined trails items-only by −0.0049. Neither figure is an artifact of the construction cohort: build and held-out curves differ by −0.0047, −0.0002 and −0.0025 across the three arms, so the test cohort scores *higher* than the cohort the sequences were built on. The criterion is prefix-optimal by construction—each arm's sequence is the best its bank can do at every budget under the same one-step rule—and that is the property an interview needs, because a question budget is a deployment choice rather than a constant. It is also the property that makes the sixteen-question column the one place a bank can be behind: a prefix-optimal path commits its opening slots on immediate gain and does not revisit them, and NDCG over a sequence is not guaranteed submodular, so containment of the item bank inside the combined bank binds at the optimum rather than along the path. Read across the table, the shape is consistent and it is the shape the rest of this section explains: the concept channel owns the opening and the tail, the item channel owns full accuracy once enough questions have been answered—and how many that is depends on how many the user can answer at all.

What the concept channel buys is answerability, and that has a literature. The figures underneath these curves are the plainest statement of the channel's value: at a realistic eight-question budget a user answers 2.18 item questions and 6.02 concept questions, a factor of 2.8. Whether a cold user *can* answer at all is not a detail of our harness—it is a first-class concern of the classical interview literature, under other names. Rashid et al. select seed items by popularity precisely because popularity is the probability that a new user has seen the item [36], and report that a pure-entropy strategy performs badly for exactly this reason; Golbandi et al.'s tree splits users three ways, into lovers, haters and *unknowns*, so unanswerability is a branch of the method rather than an afterthought [14]; and the active-learning survey of Elahi et al. gives the property its own taxonomy category, as *rateability* or acquisition probability [11]. We adopt the term *answerability* for readability, but we are renaming a known quantity, not introducing one.

The sharpest prior statement of the problem is Sun et al. [43], by three of Functional MF's own authors, whose entire contribution is an answerability fix: they report that a cold-start interview tree becomes so unbalanced that *the unknown branch captures more than* 80% *of users at each split*, so that users must repeatedly answer "unknown" before reaching a question they can engage with, and may simply abandon the interview. Their remedy is to show several items per screen, which raises the chance that a user knows at least one, and they evaluate against interview *time* rather than question count. That is the same pathology we measure, diagnosed twelve years earlier and more sharply. Two things separate our result from theirs. Their fix stays inside the item channel—it makes item questions more likely to land, rather than asking whether a different *kind* of question lands more often—and their multi-question screen is simultaneously more answerable *and* more informative, a confound they do not separate. Our comparison holds the budget and the interview protocol fixed and varies only the channel, which isolates answerability from informativeness.

We should also be clear that the *shape* of our curve is not new. Gharahighehi et al. [13] run item-only against item-plus-genre trees and find that attributes lead early while items overtake later; Xia et al. [45] find attributes dominating the opening turns of a dialogue and items the later ones. Both report the crossover we report. Neither measures an answer rate: the first attributes it to clustering efficiency, the second to preference specificity. The contribution here is therefore not the crossover but its mechanism—that the channels differ by a factor of 2.8 in how often a question is answerable at all, which the geometry of §5 predicts and which is measurable on this instrument.

On the adaptive headroom. A one-level adaptive probe on the fixed stack—branch on the first answer, then choose the second question per branch, selected and evaluated out-of-sample on disjoint halves—pays on the item channel with a CI excluding zero (+0.0015 full, CI [+0.0007, +0.0024]; +0.0016 tail, CI [+0.0008, +0.0024]), and is positive but not significant on the concept channel at one level. The effect is small because a one-level preview is not a policy; that is the point. Establishing that adaptivity pays at all, on an instrument whose ingestion is certified, is what this chapter owes the companion method chapter, whose job is to close the gap—the decision-tree acquisition line of Golbandi et al. [14] on *this* instrument and harness. The instrument chapter's job is to make the gap *measurable*; closing it is the method chapter's.

Training concepts into the tower is item-safe but buys nothing. We evaluated two homes for the signed concept fold: a trained fold-to-point on the *frozen* tower (call it the frozen-fold), and concept tokens trained *into* the tower itself (the tokens-in-tower variant). Both preserve full-profile strength, as G0 demands. The frozen-fold inherits the tower's G0 by construction (a zero-initialised fold is bit-identical on the full profile). The tokens-in-tower variant is retrained and therefore must re-earn G0, and it does: full/tail NDCG@10 0.3487/0.2471 on the test users, a paired-bootstrap tie with the 0.3540/0.2497 RecVAE bar (difference −0.0053, within the 0.0070 CI), with cold-start accuracy if anything marginally higher. So *training concepts into the tower costs essentially nothing* *on items.* It also, however, *gains* essentially nothing on concepts, and it is worse where it matters most. On per-answer concept curves the frozen-fold is higher at eight answers (0.1630/0.0639 versus 0.1566/0.0577), and in deployment currency the difference is decisive: the frozen-fold stays above the intercept through sixteen questions (0.1379 at q=16 in the ledger's paired sweep), whereas the tokens-in-tower variant, measured in the same sweep, *craters* to 0.1052, below the no-answer intercept, as the long tail of the fixed bank floods it with weak-support concepts. We therefore carry the frozen-fold as the concept operator of record; the tokens-in-tower arm is item-safe but does not earn its place. The G0 comparison above is a genuine tie on the certified ruler.

The methods lesson, restated as a battery amendment. The concept channel's near-miss is the sharpest argument in this chapter for *why the battery has the shape it does*. A capability gate (G3's sign-flip) passed on a fold that, as actually driven down a realistic question bank, never emitted a negative answer—the answer model, not the operator, was sign-blind, and the operator's certified capability was vacuous in deployment. Only the deployment-currency evaluation—a fixed, user-independent bank under a per-question budget, scored at the endpoint (the G9 ingestion-isolation setting)—exposed it, because there the declining curve penalises an answer model that never exercises the negative half. This is a general failure class, not a concept-specific accident: *a gate on what an operator* can *do must be paired with a gate on what the deployed system* does. We record the deployment-currency endpoint as the concept channel's acceptance form, and fold the signed-retrain numbers into the G3/G9 rows once the certified tower is trained (§7).

7 Experimental design

7.1 Baseline bank

The baseline bank (Table 6) is three tiers: classic floors that every curve must clear (Dacrema et al. [12] demands them), the full-profile SOTA bar that decides G0, and the elicitation-compatible rivals that decide G1/G2. Reported anchors are on the systems' *own* splits and are not comparable to ours until bridged (§7.2); we print them only to show what "ties SOTA" will mean.

7.2 Evaluation protocol

The protocol items below instantiate the headline outcome gates (the certification bridge is C1; G0 and G2 as named); they are the cheap, always-run subset. The remaining per-build gates (G3–G9) and the load-bearing answer-model-transfer certification (C2), which close the outcome-gate false positives, are specified in full in the acceptance battery (Table 1) and its design-side document, and each headline result is reported against them.

(i) Certification bridge (C1). Our ML-25M ruler and the published bar are on *different* metrics and splits, and we do not conflate them. Concretely, our paper-config RecVAE realisation reaches full/tail NDCG@10 0.3540/0.2497 on the canonical ML-25M split—whereas the published EASE/RecVAE numbers are NDCG@100 on the ML-20M strong-generalisation split. These are not comparable, and no "ties SOTA" claim is made until the bridge is run. The bridge has two steps: (1) *reproduce* the published EASE and Mult-VAE/RecVAE numbers on the ML-20M Liang split (NDCG@100, Recall@20/50) in our own code, snapping to the reported values exactly [40, 25, 38]; (2) *port* the certified tower to the ML-25M harness and report full/tail NDCG@10 there. We keep ML-20M in the loop despite being older because the Liang 2018 split is a protocol lock-in the whole full-profile literature reports on; ML-25M is the newer corpus and, crucially, carries the genome tags the concept channel needs, so both datasets are required—one for comparability, one for the channels.

The first step is complete. On the ML-20M Liang strong-generalisation split our split statistics reproduce the published counts exactly (9,990,682 events / 116,677 train users / 20,108 items), and both anchor implementations **snap to their published values**: EASE at NDCG@100 0.4203 vs. the reported 0.420 [40], and RecVAE at 0.4425 vs. the reported 0.442, with Recall@20 (0.4144 vs. 0.414) and Recall@50 (0.5525 vs. 0.553) matching as well [38]. One closed-form and one neural reproduction to the third decimal certify that our split construction, metric implementation, and both implementation families are faithful to the published protocol—every ML-25M baseline number below therefore comes from certified code.

Mult-VAE reaches NDCG@100 0.4223 against the published 0.426 [25] (−0.0037; Recall@20 0.3928 vs. 0.395, Recall@50 0.5341 vs. 0.537)—a *near*-snap, close enough to admit as a bridged Tier-2 anchor but not a third-decimal reproduction, and we label it that way rather than round it into agreement. **Mult-DAE** *fails*: 0.3969 against the published 0.419 (−0.022, with Recall@20 and @50 short by a comparable margin). We do not know whether the gap is a training-recipe detail or something deeper, and rather than carry an uncertified implementation we **drop Mult-DAE** from the baseline bank entirely—it earns no row on the ML-25M ruler.

Consistency demands we apply that rule to ourselves, and doing so costs us something. Our iALS lands at 0.358 against the published WMF anchor of 0.386—a miss of 0.028, *larger* than the 0.022 that disqualified Mult-DAE. It would be a double standard to drop one and keep the other merely because the kept row is convenient, so we state the rule and its consequence explicitly: **iALS is an uncertified floor, not a bridged anchor**. It keeps its measured row on the ML-25M ruler, where it is one more number from the same harness, but it is excluded from every claim that requires a certified comparator—no argument in this chapter rests on an unbridged number being compared with another unbridged number. Only EASE, RecVAE and (as a near-snap) Mult-VAE carry certified status on this ruler. The bifurcation argument must stand on those or not at all.

(ii) G0 — full-profile on ML-25M, items-only. On the baselines' native input (items only), evaluate full and tail NDCG@10 on the full ML-25M catalogue with a paired per-user bootstrap CI, confirming the instrument's belief mean reproduces the frozen tower and ties EASE/RecVAE on the shared harness. The item-only baselines are measured (Table 7).

A properly-trained RecVAE edges the closed-form frontier, and that is the bar. On our canonical ML-25M ruler our in-house paper-config RecVAE attains 0.3540 full / 0.2497 tail, the nominally strongest measured system—ahead of the closed-form leaders EASE (0.3476/0.2441) and EDLAE (0.3433/0.2395). The top of this ruler is a narrow band rather than a ranking: RecVAE leads EASE by +0.0064 full, about 1.8 unpaired standard errors, which is the same order as the 0.0069 paired interval that makes the tower a tie with RecVAE below. We therefore treat RecVAE as *the bar*—the strongest measured system, and the conservative one to be held to—rather than as a system separated from EASE. All rows share the same split, cohort, tail definition, and NDCG@10 routine (verified by code inspection). The point of interest for the instrument is that on this ruler a properly-trained amortised VAE—not a closed-form model—sits at the top of the frontier; the instrument's R1 obligation is precisely to *tie this bar*, reproducing RecVAE-class full/tail accuracy through the frozen tower and then preserving it under the belief layer at G0. We report the whole frontier rather than excluding the stronger baselines.

G0 is passed: the interview tower ties the bar, and is exactly the frozen decoder when the interview is empty. One certified training run of the graded-native set-encoder tower T2′—a single run of the pre-registered winning configuration, selected on validation and scored once on the held-out test cohort—reaches full/tail NDCG@10 **0.3482/0.2462** against the RecVAE bar's 0.3540/0.2497. The difference is −0.0057 full with a paired per-user bootstrap CI of 0.0069, so the two *tie* in the sense of Definition 1: the tower pays no measurable full-profile price for being interviewable. Two properties of that number matter more than its size. First, it is on the *same* ruler as every other row of Table 7—same split, cohort, tail mask, and NDCG routine—so it is a tie in the strict sense, not a bridged one. Second, the G0 *identity* half holds exactly and not merely statistically: with an empty evidence set the tower emits a zero latent ($|\boldsymbol{\mu}_0| = 0$ to numerical precision) and its scores are a rank-identical image of the frozen decoder's learned bias (Spearman 1.0000), which is what licenses reading every later interview curve as evidence about *ingestion* rather than about a shifted baseline. The cold intercept of every curve in this chapter, 0.1279/0.0192, is that same non-personalised read-out. Accuracy is therefore settled for this instrument: G0 is evaluated on the belief mean, so the tie is one the belief layer preserves rather than one it is measured beside, and the layer passes its own gates as well (§7.3). The rankings reported in this chapter are read from the posterior *mean*; putting the covariance to work in question selection is the companion chapter's subject.

7.3 The battery, measured

The battery is proposed in §2 as a portable certificate, so it owes its own results. Table 8 reports every gate.

Once a bank is answerable, which question you ask matters much less than that it is answerable. In the G2 curves, a *random* question drawn from the answerable bank beats popularity-ranked items at q=1, 2, 4, 8 (e.g. 0.2358 vs 0.2230 full at q=8) and is only overtaken at q=16 (+0.0022, CI-clean). HELF and the $\boldsymbol{\Sigma}$-greedy selector are likewise within noise of random until the largest budget, where $\boldsymbol{\Sigma}$-greedy wins by +0.0036 (CI [0.0016, 0.0057]). This is not a contradiction of G9, whose random arm is drawn from the *whole catalogue* and is therefore mostly unanswerable (mean 1.4 of 16 questions answered, versus 13.5 for the popularity bank) and which the popularity bank beats by +0.1264: the two "random" controls are different objects, and we name them separately. The lesson is that once a bank is restricted to answerable questions, *which* of them you ask matters far less than that they are answerable—which is precisely why the selection problem is deferred to the companion chapter and why we decline to claim a selection result here.

(iii) G2 — per-question curves. Cold-start per-question NDCG@10 curves (full and tail) versus a random-question control. Items are foldable by all baselines; mixed channels (concepts, signed values, continuous tokens) are foldable only by our instrument, which is the point of the comparison—the rivals cap out at what an item-only interview can carry. Symmetric access is enforced where a baseline *can* fold a channel; where it structurally cannot, that is reported as an architecture limit, not hidden. The items-only G2 curves and the G1 shrinkage diagnostic are measured (§7.3), and the mixed-channel comparison is reported in §6.4 under one strategy-free criterion. No item-only baseline appears in that comparison, because none can fold a concept at all—which is the architecture limit itself, argued in (iv) below.

(iv) Why "just add genre columns" is not an alternative. The obvious objection to the concept channel is that a concept could simply be appended to an item-only model as an extra column—a centroid-of-members pseudo-item in EASE, or an extra input unit in RecVAE. The objection fails for a structural reason rather than an empirical one, and the reason is worth stating precisely because it is the same property that motivates the whole channel.

A column is fixed when the model is fitted. EASE's item–item weight matrix is defined over a fixed column set and is obtained in closed form from the Gram matrix of exactly those columns; RecVAE's input layer has one weight vector per column. Adding a concept therefore means adding a column, and adding a column means *refitting the model.* This is not a tuning inconvenience: it means the set of expressible concepts is frozen at training time. A paraphrase of an existing concept, a concept invented after deployment, a direction produced by snapping a continuous query to the nearest meaningful axis—none of these can be answered without retraining, because none of them has a column. The pseudo-item is not merely a lossy encoding of a concept; it is a *closed* one.

Metrics. Full *and* tail NDCG@10 on ML-25M (the project reporting rule, scored with the decoder's learned bias, not a log-count popularity floor), plus NDCG@100 (and Recall@20/50) on the ML-20M bridge split for comparability with the published anchors. Every headline delta carries a paired per-user bootstrap CI on one shared harness. Every cell reported is measured on this harness.

8 Limitations and threats to validity

- **Metric non-comparability until the bridge is run.** Our frontier number 0.3540/0.2497 is NDCG@10 on ML-25M; the published bar is NDCG@100 on ML-20M. Until §7.2 snaps our code to the published numbers (done for EASE/RecVAE, §7.2), *no* "ties SOTA" claim across the two rulers is licensed—the ML-25M and ML-20M numbers live on different rulers.
- **Monotonicity is empirical, not a theorem.** Conjugate-Gaussian updates are monotone in expected utility [15, 6] *only in expectation*, and only if the ranker acts optimally under the belief—which a fixed dot-product / NDCG ranker does not. R5 (and G2's rising curve) is therefore a *measured* property with a per-question CI, not a guarantee; a truthful answer can in principle still lower NDCG at a given step, and we report any such step rather than suppress it.
- **Answers come from recorded behaviour.** G1/G2 are measured under the answer model that certification C2 fixes: an item answer is the user's own recorded rating and a concept answer is a log watch-lift over raw counts and item marginals, so no recommender-derived quantity can enter an answer and the self-simulator inversion is closed by construction rather than by measurement. These are real preferences, but they are not a person responding to a rendered question, and the chapter makes no claim about how a user would phrase, hedge or misread one.
- **BERT4Rec replicability.** The sequence baseline's fold-in advantage is subject to a documented replicability caveat [34]; we tune it to the published protocol and flag any residual gap rather than treat a single run as authoritative.
- **Graph-filter admissibility.** GF-CF/BSPM/Turbo-CF [37, 8, 32] do not publish an ML-20M strong-generalisation NDCG, so no row in this family can be a *bridged* tie. Turbo-CF is re-run on our split with (α, s, filter) selected on our validation users and is reported as best-effort tuned; GF-CF and BSPM are not run here and carry no number.
- **Scope of the novelty claim.** The conjunction claim is bounded by our survey: a set-encoder / deep-set recommender that reached RecVAE-class full-profile accuracy *and* took value-carrying

mixed-type tokens would partly pre-empt R1$\wedge$R2$\wedge$R3, and we found no such reported system.

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Gate	One-line criterion	Failure class it excludes
Per-build gates (every build; all cheap)		
G0	Belief mean reproduces the frozen tower's full/tail NDCG@10 (tie or bit-identity).	Full-profile regression from the belief layer.
G1	$\mathbf{\Sigma}$ shrinks in expectation over answers (refusals exempt), and shrinks directionally along the queried $\boldsymbol{\phi}$.	Unusable posterior; scale-only covariance.
G2	Cold full/tail curve strictly rises to budget; a single answer from cold lifts both channels.	Flat-model and out-of-envelope artifacts.
G3	Sign-flip lowers NDCG; value-neutralisation costs NDCG; quality monotone in intensity.	Sign-blind and intensity-blind folds.
G4	A refusal burns the turn and does not move the belief (architectural invariant).	Unanswerable questions silently scored as evidence.
G5	Concept lifts member items (AUC $>$ 0.8) and clears the no-answer intercept from cold.	Non-specific / popularity-parasite concepts.
G6	Wrong-user, shuffled, placebo, and duplicated answers gain nothing over the true-answer arm.	Cardinality confound; existential leakage.
G8	Replacing $\mathbf{\Sigma}$ with $c\mathbf{I}$ degrades the deciding number at budget; order- and re-ask-invariant.	Decorative $\mathbf{\Sigma}$; order-dependence.
G9	On a fixed user-independent question set the curve still rises from answer content.	Taste-leak: selection masquerading as ingestion.
One-time certifications (per instrument)		
C1	Implementations snap to published numbers on the canonical split (met; Mult-DAE failed and was dropped).	Uncertified ruler / metric mismatch.
C2	Answers are computed from observed behaviour only; no recommender-derived quantity enters an answer (asserted in code).	Circular answer model (self-simulator inversion).

Table 1: **The instrument acceptance battery.** Per-build gates G0–G9 (there is no standalone G7; the answer-model-transfer check is folded into C2) run on every build; certifications C1–C2 are one-time per instrument. The three headline outcome gates are G0/G1/G2; the semantic and artifact-control gates G3–G9 and the transfer certification C2 close the false positives that pass the outcome gates alone (text). The full specification, with each gate's testable form, is the design-side battery document.

Family	Representatives	R1	R2	R3	R4	R5	One-line verdict
(i) retrain / meta-learn per user	MeLU, TaNP [26], CDRNP	∼	✓	✗	∼	✗	R2/R4 precedent; not full-profile SOTA, not channel-open.
(ii) fold-in linear / closed-form	EASE [40], SLIM, GF-CF [37], BSPM [8], Turbo-CF [32]	✓	✓	✗	✗	∼	The full-profile bar (G0); item-indicator basis ⇒ R3 impossible.
(iii) amortised inference encoders	Mult-VAE [25], RecVAE [38], EDLAE [41], EDDI [30]	✓	✓	∼	∼	✗	The architecture we live in; VAEs item-only, EDDI right shape/wrong domain.
(iv) Bayesian belief over item space	Bıyık [5], ConTS [23], BCIE [44], ICF [46]	✗	✓	∼	✓	∼	The R4/R5 heart; belief layers, not full-profile towers.
(v) sequence models	SASRec [20], BERT4Rec [42]	∼	✓	✗	✗	✗	Interview ≠ ordered session; tokens item-only, order-sensitive.
(vi) LLM-as-recommender / CRS	PEBOL [3], GATE [22], LLM-CRS [17]	✗	✓	✓	∼	✗	Best R3 (open text); weakest R1 (weak collaborative signal).
CASPER (target)	**this chapter**	✓	✓	✓	✓	✓	**The conjunction; no published system occupies it.**

Table 2: Six-family taxonomy scored against R1–R5. ✓ met, ∼ partial, ✗ not met. Scores are our reading of each family's *published* claims and are qualitative; the last row is a target, not a result. R1 and {R3,R4,R5} are anti-correlated: families (ii)/(iii) hold the accuracy corner, family (iv) the belief corner, and none spans both.

(a) Classic / linear full-profile								
Most-Popular	✗	✓	✗	✗	∼	.1345 / .0226	measured	n/r (non-personalised floor)
Item-kNN (cosine)	✗	✓	✗	✗	∼	.2428 / .1297	measured	@100 ∼ 0.36 class [9]
PureSVD / iALS-WMF	✗	✓	✗	✗	∼	.2442 / .1853	measured (iALS)	@100 0.386 (WMF, ML-20M)
EASE [40]	✓	✓	✗	✗	∼	.3476 / .2441	measured	@100 0.420 (ours **PASS** +.0003)
EDLAE [41]	✓	✓	✗	✗	∼	.3433 / .2395	measured	@100 ≈ 0.42–0.44 class
Turbo-CF [32]	∼	✓	✗	✗	∼	.2622 / .1621	measured (val-tuned)	n/r (ML-20M not reported)
GF-CF / BSPM [37, 8]	∼	✓	✗	✗	∼	—	not run here	n/r (ML-20M not reported)
(b) Amortized / neural full-profile								
Mult-DAE [25]	✓	✓	✗	✗	✗	—	*dropped* (snap failed)	@100 0.419 (ours 0.397, **FAIL** −.022)
Mult-VAE [25]	✓	✓	✗	∼	✗	.3202 / .2194	measured	@100 0.426 (ours 0.422, near-snap −.004)
RecVAE [38] (paper config; our run)	✓	✓	✗	∼	✗	**.3540** / **.2497**	measured	@100 0.442 (impl. **PASS** +.0005)
SASRec / BERT4Rec [20, 42]	∼	✓	✗	✗	✗	—	not run here	n/r (next-item; repl. [34])
TaNP [26] (MeLU / CDRNP line)	∼	✓	✗	✓	✗	—	not run here	n/r (own cold-start benchmark)
(c) Elicitation / CRS systems — the belief corner								
EDDI / Partial-VAE [30]	✗	✓	∼	✓	✗	—	not run here	n/r (non-recsys; UCI/MNIST)
Golbandi tree [14]	✗	✓	✗	✗	✗	.3065 / .1895	measured (core)	n/r (Netflix RMSE)
RBMF / Functional-MF [27, 47]	✗	✓	✗	✗	✗	.2534 / .1764	measured (best-effort)	n/r (elicitation RMSE)
Bıyık soft-attributes [5]	✗	✓	∼	✓	∼	—	reports no full-catalogue accuracy	n/r (16 users, slates of 5)
ICF [46]	✗	✓	✗	✓	∼	—	core ≈ iALS (measured)	n/r (online CF; own protocol)
ConTS [23]	✗	✓	∼	✓	∼	—	reports no full-catalogue accuracy	n/r (cold-start CRS metrics)
BCIE [44]	✗	✓	∼	✓	∼	—	not portable (needs their KG)	n/r (critique metrics)
UNICORN [10]	✗	✓	∼	✓	✗	—	not portable (needs their KG)	n/r (graph-RL CRS; own metrics)
PEBOL [3]	✗	✓	✓	∼	✗	—	not runnable at full catalogue	MRR@10 0.27 vs 0.17 (cold-start only)
GATE [22]	✗	✓	✓	✗	✗	—	not runnable (no ranker)	n/r (elicitation-only; no full ranking)
LLM zero-shot CRS [17]	✗	✓	✓	✗	✗	—	not runnable at scale	n/r (weak collaborative signal)
(d) Ours								
Set-encoder tower T2′ (ours)	✓	✓	∼	∼	∼	.3482 / .2462	**measured (certified)**	n/a (in-house; graded-native tower)
Instrument (target)	✓	✓	✓	✓	✓	—	pending	n/a (this chapter)

Table 3: **Master table: every reasonable system by R1–R5 and by measured full-profile accuracy on one shared ruler.** ✓ met, ∼ partial, ✗ not met (our reading of each system's *published* claims, consistent with Table 2). *full/tail NDCG@10* is on the canonical Liang-recipe strong-generalization ML-25M split (catalog restricted to the 18,430-item project catalog, train vocabulary 18,359; train 140,768 / val 10k / test 10k users; per-user 80/20 fold-in; tail = top-33%-train-mass head mask, 190 head items), scored with the decoder's learned bias. The Bıyık and ConTS rows carry a dash because neither reports full-catalogue accuracy; the Golbandi row prints its node-recommender core. Statuses are *measured* (run on this ruler; best-effort = with stated substitutions), *not run here* (runnable in principle, no number claimed), or *not runnable / not portable* (impossible by construction; mechanism in the text). The final column is each system's own published full-profile anchor on its own split/metric—*not* comparable to our ruler until bridged (§7.2); "n/r" where the literature reports none. The RecVAE row is our own paper-config training run on this ruler and is the current full-profile frontier; the "ours" block adds the graded-native interview tower T2′, whose certified run ties that frontier (0.3482/0.2462; §7). Mult-DAE carries a

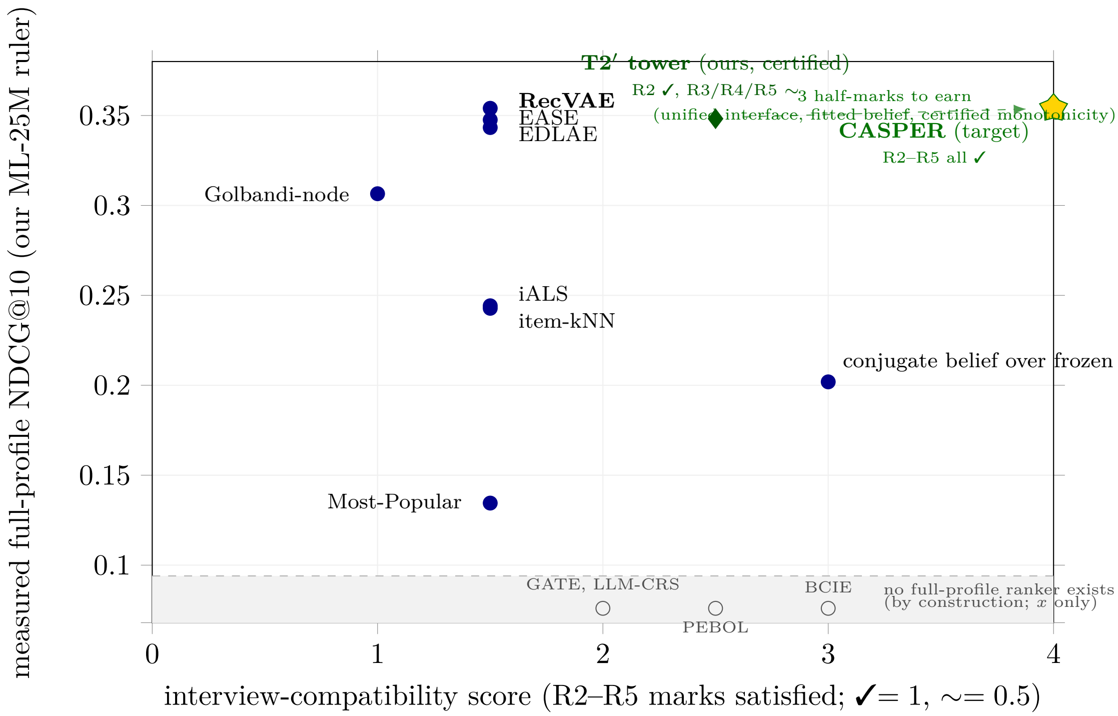

Figure 1: **The gap map (measured axes).** y-axis: measured full-profile NDCG@10 on our canonical ML-25M ruler. x-axis: an *interview-compatibility score*, defined as the number of **R2–R5** marks a system carries in the master table (Table 3), scoring ✓=1 and ∼=0.5, maximum 4. *R1 is deliberately excluded from the score*: accuracy is the y-axis, and letting it also inflate x would blur the very trade-off the figure exists to show. The score is a *coarse ordinal capability count, not a measurement*, so the x-axis is qualitative even though the y-axis is measured. Filled markers are systems with a full-profile number on this ruler; Golbandi-node is that tree's best-effort node recommender, and the conjugate-belief point is our own posterior-mean ranker (§5), plotted to show what ranking on a conjugate mean costs. Hollow markers in the separated grey strip *below* the measured y-scale are systems that *cannot produce a full-profile number by construction* (no full-catalogue ranker, or not portable; see the master-table dispositions); only their x position is meaningful. Runnable-but-not-yet-measured systems (sequence models, graph filters, EDDI, TaNP) are *absent* from the figure and join the plot when their numbers land. The star marks the instrument target (all five requirements at the frontier bar); the green diamond is our *certified* tower T2′ at 0.3482—the first measured point that is simultaneously at the frontier bar and away from the item-only column. T2′ sits at x=2.5: it holds R2 outright, and holds R3, R4 and R5 at a half-mark each, since concepts fold through the trained operator of §6 with the open-vocabulary path illustrated rather than deployed; the covariance is maintained analytically and gated (G1a, G1b) rather than used to rank, which §5 shows is the right choice on this ruler; and per-question monotonicity is measured with a CI rather than guaranteed. The population is otherwise *bimodal*: the entire accuracy corner (Pop through RecVAE, 0.343–0.354 at the top) collapses onto $x \leq 1.5$—strong recommenders that can fold items and nothing else—while every system with genuine interview machinery sits at $x \geq 2$ and reports no full-catalogue accuracy at all. The one point plotted at x=3 is not a rival but our own conjugate posterior-mean ranker at 0.2019, included because it shows what the uniform-fold alternative costs. Two clusters, no bridge: the corner is not expensive; it is unoccupied. The nearest published rival to the target is Bıyık et al. [5], the highest compatibility of any system with an actionable belief.

free-text phrase	nearest concept	cos	top of the returned ranking
"the quiet dignity of losing"	*redemption*	0.47	Umberto D.; Tyson; Red Army; Days of Glory
"people trapped by a decision they made years ago"	*history*	0.39	Music Box; Judgment at Nuremberg; Missing
"two people talking for the whole film"	*dialogue*	0.54	Sweet Smell of Success; Trouble in Paradise
"movies about grief"	*feel good movie*	0.54	Rachel, Rachel; Mean Creek; The Woodsman

Table 4: **A phrase with no curated counterpart is placed, not looked up.** Four free-text phrases, each screened against the 1,031-concept vocabulary two ways—no genome tag occurs as a whole-word substring, and no concept sits within cosine 0.6—with the nearest concept we actually hold printed so the distance from anything curated is visible. Directions come from the single global map fitted on 80% of the concepts; nothing here is retrained or looked up.

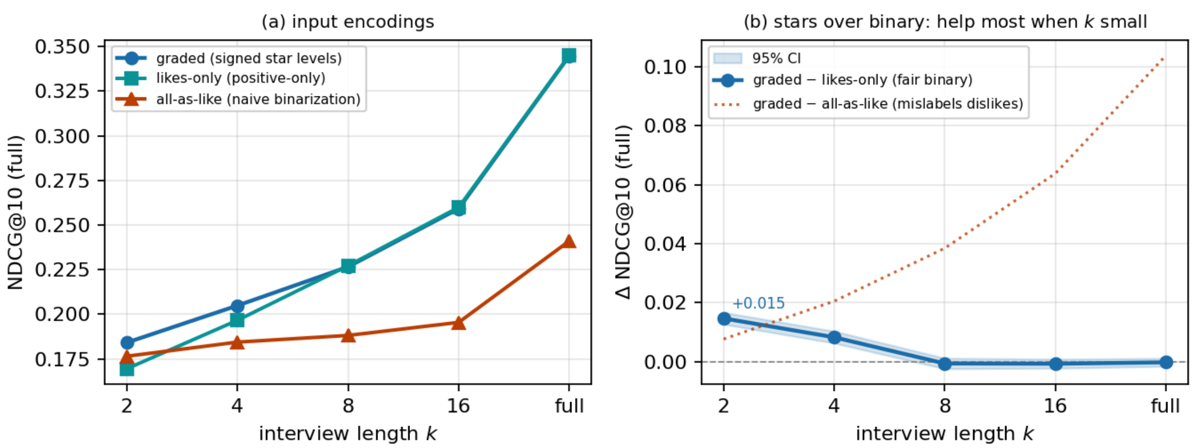

Figure 2: **Graded input helps over binarized input, most on the short interview** (ML-25M interview harness, full NDCG@10 vs. interview length k). **(a)** The same revealed set is fed three ways: *graded* (real signed star levels), *likes-only* (sub-threshold answers dropped—exactly what a positive-only tower such as RecVAE does at $r>3.5$), and *all-as-like* (every answered item recorded as a 4-star like—naive implicit binarization). **(b)** Against the fair positive-only baseline, the graded premium is *CI-clean positive on the short interview*—+0.0147 at $k=2$ (CI $[+0.0127, +0.0167]$), $+0.0083$ at $k=4$ (CI $[+0.0063, +0.0103]$)—and fades to a tie from $k=8$ onward (-0.0006, CI $[-0.0024, +0.0012]$) as collaborative signal fills in: the extra bits a star carries are most decisive when few items are known. The *mechanism* is that a signed answer encodes a dislike (a negative y_j the item-indicator basis cannot express); against the *all-as-like* control, which mislabels dislikes as likes, the apparent premium instead *grows* with k (dotted)—a control artifact (more revealed items $\Rightarrow$ more dislikes corrupted), not value from gradation. Mirroring the star signs costs -0.2985 on the full-profile arm of G3a (Table 8).

bank	q=1	q=2	q=4	q=8	q=16	answered of 8
items-only	.1405/.0275	.1524/.0387	.1656/.0445	**.1863**/.0538	**.2098**/.0679	2.18
concepts-only	**.1436**/**.0323**	.1504/**.0434**	.1647/**.0551**	.1776/**.0698**	.1808/.0758	**6.02**
combined	**.1436**/**.0323**	.1504/**.0434**	.1647/**.0551**	.1817/.0616	.2049/**.0776**	4.67

Table 5: **Best static question sequences, one strategy-free criterion per bank.** Full/tail NDCG@10 at each budget, sequences built by lazy greedy on the 10,000 validation users over a 500-item and/or 500-concept bank and scored once on the 10,000 disjoint test users; no refusals; intercept 0.1279/0.0192. The last column is the mean number of the first eight questions a user can actually answer. Bold marks the best bank in each column. Paired per-user bootstrap intervals on 10,000 users at this scale have half-widths of ±0.002–0.003 throughout this chapter; the differences read in the text are the ones comfortably above that.

Baseline	Type	Reported anchor	Why the slot
Tier 1 — classic floors (must clear)			
Most-Popular	non-personalised	popb floor (our harness)	the "did we learn anything" line; the G2 curve must clear it at budget.
Item-kNN (cosine)	neighbourhood	ML-20M NDCG@100 ~ 0.36 class	instant fold-in; Cremonesi-era strong-simple [9].
PureSVD / iALS-WMF	linear MF fold-in	WMF ML-20M NDCG@100 0.386	cheap fold-in reference.
SLIM	sparse linear item-item	ML-20M NDCG@100 0.401	direct EASE ancestor.
Tier 2 — full-profile SOTA (the R1 / G0 bar)			
EASE [40]	closed-form item-item	NDCG@100 0.420	the bar to tie; item-only shows the R3 gap.
RecVAE [38]	amortised VAE	NDCG@100 0.442	top of the ML-20M split; our in-house frontier.
Mult-VAE [25]	amortised VAE	NDCG@100 0.426 (ours 0.422)	canonical amortised anchor; near-snap.
Mult-DAE (dropped)	DAE	NDCG@100 0.419 (ours 0.397)	snap **failed** -0.022; disqualified.
EDLAE [41]	denoising higher-order EASE	$\approx$ RecVAE class	confirms the flat frontier.
GF-CF / BSPM / Turbo-CF [37, 8, 32]	closed-form graph filter	SOTA on Gowalla/Yelp (ML-20M not reported)	admit only if re-run on our split, which Turbo-CF now is: 0.2622/0.1621, val-tuned.
SASRec / BERT4Rec [20, 42]	self-attentive sequence	strong next-item (not this split)	sequence fold-in; watch replicability [34].
Tier 3 — elicitation-compatible rivals (decide G1/G2)			
EDDI / Partial-VAE [30]	set-encoder + info-gain	non-recsys (UCI/MNIST)	the R2 architecture; myopic info-gain baseline.
Bıyık soft-attributes [5]	Gaussian belief + EVOI	own elicitation task (16 users, slates of 5)	the central threat; open-vocabulary and R1-bar wedge.
ConTS [23]	Thompson bandit, item+attr arms	cold-start CRS metrics	unified-space + uncertainty; fixed schema.
BCIE [44]	conjugate-Gaussian critique fold	critique metrics	exact-conjugacy-over-frozen precedent.
PEBOL [3]	Bayesian-opt NL-PE (LLM)	MRR@10 0.27 vs 0.17	load-bearing PE baseline; per-item Beta.
Golbandi tree [14]	RMSE decision-tree interview	Netflix RMSE	the static-tree interview comparator.
RBMF / Functional-MF [27, 47]	MF-embedding seeds	elicitation RMSE	interview-seed line; cite-and-differentiate.

Table 6: Baseline bank. Anchors are on each system's own split/metric and are *not* comparable to ours until the bridge (§7.2) is run; G0 is decided against EASE+RecVAE, G1/G2 against the Tier-3 rivals plus a random-question control. Any deep-net "SOTA" not snapping to Tier-2 numbers on our split is inadmissible [12].

Method	full NDCG@10	tail NDCG@10	note
Most-Popular	0.1345	0.0226	@100 0.1975; non-personalised floor
conjugate belief over frozen MF	0.2019	0.1200	@100 0.2886; the posterior-mean ranker (§5)
Item-kNN (cosine)	0.2428	0.1297	@100 0.3200; neighbourhood fold-in
iALS-WMF	0.2442	0.1853	@100 0.3704; MF fold-in
RBMF / Functional-MF seed	0.2534	0.1764	@100 0.3607; best-effort seed core
Turbo-CF [32]	0.2622	0.1621	@100 0.3569; val-tuned α=0.5, s=1, 2nd-order
Golbandi node recommender	0.3065	0.1895	@100 0.3958; best-effort tree core
Mult-VAE [25]	0.3202	0.2194	@100 0.4303; near-snap −0.004
EDLAE [41]	0.3433	0.2395	@100 0.4330; val p=0.4
EASE [40]	0.3476	0.2441	@100 0.4375
RecVAE (paper config; our run) [38]	**0.3540**	**0.2497**	@100 0.4537; nominal frontier
Ours			
Set-encoder tower T2′ (certified)	0.3482	0.2462	@100 0.4486; **G0 tie** (−0.0057, CI 0.0069)

Table 7: G0 — full-profile, items-only, on the canonical Liang-recipe strong-generalization ML-25M split (catalog restricted to the 18,430-item project catalog, train vocabulary 18,359; train 140,768 / val 10k / test 10k users; per-user 80/20 fold-in; seed 98765), NDCG@10 (full and tail); tail = top-33%-train-mass head mask (190 head items); scored with the decoder's learned bias. The belief-MF, RBMF and Golbandi rows are best-effort cores of the belief-corner systems (§3). All rows reuse the canonical harness verbatim, so they are on one ruler.

Gate	Criterion	Measured	Verdict
G0	tie the bar; identity at t=0	0.3482/0.2462 vs 0.3540/0.2497, diff -0.0057 (CI 0.0069); $\|\boldsymbol{\mu}_0\|$=0, Spearman 1.0000 vs decoder bias	**PASS**
G1a	tr $\boldsymbol{\Sigma}$ shrinks over answers	1881.9→1805.1 across q=0 . . . 16, monotone	pass
G1b	directional, $\geq 2\times$	along-$\boldsymbol{\phi}$ 0.996 vs elsewhere 0.086 = **11.5**×	pass
G2‡	cold curve rises, beats random	$\boldsymbol{\Sigma}$-arm 0.1279 → 0.2724 full, 0.0192 → 0.1305 tail; rise +0.1445/ + 0.1113. Canary: one answer from cold rises on *both* channels, full and tail (items 0.1405/0.0275, concepts 0.1436/0.0323)	**PASS**‡
G3a	sign-flip lowers NDCG	full-profile arm: 0.4279 → 0.1294, -0.2985 (CI $[-0.3047, -0.2922]$); intensity staircase $\rho = 0.927$	**PASS**
G3b	value-neutralisation costs	full-profile arm: 0.4190 → 0.1321, drop 0.2869 (CI $[0.2817, 0.2922]$); tail 0.2999→0.0368	**PASS**
G4	refusals inert (architectural)	unanswerable questions are never folded; $\alpha_{\text{refuse}}/\alpha_{\text{answer}} = 0.0007$	**PASS**
G5	concept lifts members, beats its popularity projection	whitened-direction membership AUC 0.94 (raw 0.44); cold fold clears the intercept 0.1308 > 0.1279	**PASS**
G6	wrong-user / shuffle / placebo / duplicate gain nothing	true arm beats shuffled by +0.0371 and placebo by +0.0168 (both CI-clean); duplicate assert fired; wrong-user -0.0986 (a stranger's taste is worse than silence)	**PASS**
G8	$\boldsymbol{\Sigma}$ beats isotropic $c\mathbf{I}$ at budget; order-invariant	+0.0089 at q=16 (CI $[0.0071, 0.0107]$), also wins q=4, 8; order deviation 5×10^{-7}	**PASS**
G9	fixed bank rises from answer content	rise +0.1218 (CI $[0.1184, 0.1252]$), monotone $\rho = 1.0$	**PASS**
C1	snap to published numbers	EASE +.0003, RecVAE +.0005, Mult-VAE −.004; **Mult-DAE** −.022 **failed, dropped**	**PASS**†
C2	answers carry no recommender geometry	item answer = own rating; concept answer = log watch-lift over raw counts	**by construction**

Table 8: **The acceptance battery, measured.** We gate the covariance on its *structure* (G1a, G1b) and not on the absolute scale of its uncertainty: a control shows that the scale question is confounded by target difficulty—the mean popularity of a user's held-out items predicts their NLL with $|\rho| = 0.70$, where the belief's own σ reaches 0.24—so a scale-calibration gate would largely measure which users happen to have easy targets. G1b, at 11.5× against a 2× bar, is the load-bearing evidence that the covariance carries per-direction information. G2 is measured on a bank filtered per user to the questions they can answer, so it isolates *ingestion* from question *availability*; a deployed static sequence faces the availability problem separately (§6). G3a and G3b are *full-profile* corruption arms—the user's whole rated history is folded and then flipped or neutralised—so their uncorrupted references (0.4279 and 0.4190 full, 0.2999 tail) sit above the 80/20 fold-in values of Table 7, which withhold a fifth of each profile as targets. The gate is the *drop* within an arm; references are never compared across arms. There is no standalone G7: the answer-model-transfer check is folded into certification C2. †C1 covers EASE, RecVAE and Mult-VAE (§7.2).